DATE:25/08/2026

NAME:P.Moshiya

REG NO:192572142

EXPERIMENT 27

PROLOG PROGRAM – BEST FIRST SEARCH ALGORITHM

Aim

To implement the **Best First Search algorithm** using Prolog to find a path from a starting node to a goal node based on heuristic values.

Algorithm

1. Define the graph using edge/2 facts.
2. Define the heuristic value of each node using h/2.
3. Start the search from the given starting node.
4. Store the current node and its path in the OPEN list.
5. Select the node having the **lowest heuristic value**.
6. If the selected node is the goal, display the path.
7. Otherwise, generate its neighboring nodes.
8. Add the neighboring nodes to the OPEN list.
9. Repeat the process until the goal is found or the OPEN list becomes empty.

Prolog Code

```
% Best First Search in Prolog


% Graph

edge(a, b).

edge(a, c).

edge(b, d).
```

```
edge(b, e).
edge(c, f).
edge(c, g).
edge(d, h).
edge(e, h).
edge(f, i).
edge(g, i).

% Heuristic values
h(a, 6).
h(b, 4).
h(c, 3).
h(d, 5).
h(e, 2).
h(f, 4).
h(g, 1).
h(h, 1).
h(i, 0).

% Best First Search
best_first(Start, Goal, Path) :-
    search([[Start]], Goal, RevPath),
    reverse(RevPath, Path).
```

```prolog
% Goal reached
search([[Goal|Rest]|_], Goal, [Goal|Rest]).

% Continue search
search([[Node|Path]|Open], Goal, Solution) :-
   findall(
      [Next,Node|Path],
      (edge(Node, Next),
       \+ member(Next, [Node|Path])),
      Children
   ),
   append(Open, Children, NewOpen),
   sort_nodes(NewOpen, SortedOpen),
   search(SortedOpen, Goal, Solution).

% Sort paths according to heuristic value
sort_nodes([], []).

sort_nodes([Path|Paths], Sorted) :-
   sort_nodes(Paths, Sorted1),
   insert_path(Path, Sorted1, Sorted).

% Insert path according to heuristic value
insert_path(Path, [], [Path]).
```

```
insert_path([Node|_]=Path, [[Node2|_]=Path2|Rest],

        [[Node|_]=Path, [Node2|_]=Path2|Rest]) :-

   h(Node, H1),

   h(Node2, H2),

   H1 =< H2,

   !.


insert_path(Path, [Path2|Rest], [Path2|Sorted]) :-

   insert_path(Path, Rest, Sorted).
```

Input

best_first(a, i, Path).

Output

```
| ?- consult('C:/Users/MOSHIYA/Downloads/expp 27.pl').
compiling C:/Users/MOSHIYA/Downloads/expp 27.pl for byte code...
C:/Users/MOSHIYA/Downloads/expp 27.pl compiled, 63 lines read - 5585 bytes written, 16 ms

(16 ms) yes
| ?- best_first(a, i, Path).

Path = [a,c,g,i] ?
```

Result

Thus, the **Best First Search algorithm was successfully implemented in Prolog**, and the path from the starting node a to the goal node i was obtained as:

[a,c,g,i]